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Enhanced charge transport in nano-confined ionic liquids

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Charge transport in ionic liquids contained in unidirectional nanoporous membranes (pore diameters: 7.5–10.4 nm) is investigated by combining broadband dielectric spectroscopy (BDS) and pulsed field gradient (PFG)-NMR. This enables one to determine the diffusion coefficient and the diffusion rate over more than 13 decades and to trace its temperature dependence. Under conditions of nanometric confinement, a change from a Vogel–Fulcher–Tammann into an Arrhenius-like thermal activation is observed, resulting in an enhancement of diffusivities by more than two orders of magnitude. The effect becomes more pronounced with decreasing pore diameter. It is attributed to changes in molecular packing and hence in density leading to higher mobility and electrical conductivity.

Introduction

Charge transport and molecular dynamics in confined amorphous ion-conductors are topics of fundamental as well as technological importance. Ionic liquids form a special category of ion-conductors consisting entirely of ions. Despite the existence of reports as early as 1914,¹ significant research interest in ionic liquids (ILs) was only awakened in the past decade following the discovery of the unique mix of desirable features such as low melting points, low vapour pressures, wide liquid ranges, high thermal stability, tunability, high electrical conductivity as well as wide electrochemical windows. They are under investigation for uses in batteries and supercapacitors, solar and fuel cells, electrochemical deposition of metals and semiconductors, protein extraction and crystallization, nanotechnology applications, physical chemistry, and many others.^{2–11} Recent experimental studies of certain ILs under nano-confinement have demonstrated reduction of molecular mobility and conductance. For instance, using specialized nuclear magnetic resonance (NMR) techniques, Le Bideau *et al.* found a decrease in the diffusivity of ILs in monolithic silica matrices.¹² Davenport *et al.* observed about three-fold reduction in the conductance of ILs in small single-pores (<10 nm) and explained their results in terms of the “hindered transport theory”.¹³ Recently, we demonstrated that the decrease in diffusivity of a selected IL in directed silica mesopores (7.5 nm pore diameter) could be reversed by silanization.¹⁴ These reports emphasize the significance of the pore sizes, pore wall–IL interaction, as well

as the physico-chemical properties of the IL investigated. Whereas experimental results have shown slower dynamics in nanopores, theoretical methods have predicted both slower and faster dynamics of ILs in nanopores depending on the molecular details of the confining space as well as the IL. For example, Wang *et al.* performed classical molecular dynamics simulations of confined ILs in contact with graphite surface and noted remarkable adsorption and formation of dense double-layers leading to reduction of the ionic mobility. Conversely, while investigating sorption of CO₂ and H₂ into ILs confined in carbon nanotubes by a combination of molecular dynamics and Monte-Carlo simulations, Shi and Sorescu found an increase of self-diffusion coefficients by 1–2 orders of magnitude in comparison to corresponding bulk IL values.¹⁵ No experimental results confirming this prediction have been reported for confined ionic liquids.

In the current work, we present the first experimental evidence of enhancement of molecular mobility and ionic conductivity of a confined ionic liquid by 2 orders of magnitude in nanopores. We employ broadband dielectric spectroscopy (BDS) and pulsed field gradient NMR (PFG NMR) to study diffusivity of IL spanning more than 12 orders of magnitude in a broad temperature range.

The IL studied in this work (1-butyl-3-methylimidazolium tetrafluoroborate—BMIM BF₄, molecular weight = 226.024 g mol^{−1}, purity >99%) was purchased from Iolitec GmbH. The IL was dried in vacuum (10^{−6} to 10^{−7} mbars) at 60 °C for 12 hours prior to measurements. Porous silicon membranes were prepared by electrochemical anodic etching of highly doped p-type (100) oriented silicon substrates with low resistivities (<0.005 Ω cm). The etching parameters determine the mean pore sizes, geometry of the pores as well as the thickness of the exfoliated silicon membranes. In this study, we used porous SiO₂ membranes (the procedure for oxidation and filling-in of membranes with ILs is given in our previous article¹⁴) of about 50 ± 2 μm thickness and mean pore diameters of 7.5, 9.5 and 10.4 nm.

In order to characterize the silica nanopores, scanning electron microscopy (SEM) and NMR-cryoporometry were employed¹⁶ (see Fig. 1(a) and (b)). To suppress the formation of an interfacial layer due to the presence of dangling hydrogen bonds, the inner surfaces of the nanoporous silica membranes were modified by silanization using hexamethyldisilazane (HMDS) with purity >99%, purchased from Sigma Aldrich GmbH. The silanol groups were replaced by trimethylsilyl groups and the inner surfaces become hydrophobic as confirmed by the resulting infrared (IR) spectra.¹⁴ For the IR measurements, a Varian FTIR-Spectrometer equipped with a UMA

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500 microscope working in transmission mode was employed. A photovoltaic mercury-cadmium-telluride detector (KMPV50, Kolmar Technologies) operational in the range 750–5000 cm^{-1} with a resolution of 1 cm^{-1} was used. A Linkam THMS600 heating stage with a TMS91 controller and ZnSe windows was utilized for temperature dependent studies (133–373 K). Fig. 1(c) shows IR absorption spectra for the SiO_2 membrane in bulk BMIM BF_4 as well as in silica nanopores at 300 K. The peaks in the spectral region 2750–3000 cm^{-1} as well as 1335–1341 cm^{-1} are assigned to the CH_2 stretching vibrations of the alkyl chain and C–N imidazolium ring vibrations, respectively.^{17–19} To gain further insight into the morphological changes in the IL, the temperature-dependent IR spectra of the IL are studied. It is well established that the frequency of C–N stretching vibration increases with effective density.^{17–19} The results presented in Fig. 1(c) indicate a decrease in the density of BMIM BF_4 under confinement. The frequencies in bulk and confined IL coincide within the limits of experimental accuracy above the calorimetric glass transition temperature, T_g , but show deviations near and below T_g . The lower frequency in the case of the confined IL implies that at about $T < 210$ K the density is slightly lower than bulk IL, probably due to steric effects in the nanopores.

Broadband dielectric measurements were carried out using a Novocontrol high resolution alpha dielectric analyzer (0.1 Hz–10 MHz) and a HP impedance analyzer (1 MHz–1.8 GHz) in the

temperature range 170 to 400 K, stabilized to better than 0.1 K by Quatro temperature controllers using pure nitrogen as the heating agent. The dielectric and conductivity spectra are corrected to eliminate the contribution of the porosity of the membranes to the measurements. A 400 MHz NMR spectrometer with a home-built gradient unit was employed to obtain the self-diffusivities at different temperatures.²⁰ The stimulated spin echo pulse sequence with observation time of 100 ms was used. The spin echo attenuation showed a mono-exponential behaviour, *i.e.* one-component diffusion over the whole temperature range examined.

Due to its ability to probe molecular fluctuations and charge transport in broad frequency and temperature ranges, broadband dielectric spectroscopy (BDS) turns out to be an especially versatile experimental tool for probing charge transport in ionic liquids.^{21–23} BDS measures the complex dielectric function, ϵ^* ($=\epsilon' - i\epsilon''$), as well as the complex conductivity function, σ^* ($=\sigma' + i\sigma''$). At low applied electric fields (within the linear response regime), the two functions are given by $\sigma^*(\omega, T) = i\epsilon_0\omega\epsilon^*(\omega, T)$, implying that $\sigma' = \epsilon_0\omega\epsilon''$ and $\sigma'' = \epsilon_0\omega\epsilon'$ where ϵ_0 and ω denote the permittivity of vacuum, and radial frequency, respectively. For ion-conducting amorphous materials, the dielectric spectra are also often presented in terms of the complex modulus, M^* ($=1/\epsilon^* = i\omega\epsilon_0/\sigma^*$), with the real and imaginary parts of the complex modulus given by $M' = \epsilon_0\omega\sigma''/((\sigma')^2 + (\sigma'')^2)$ and $M'' = \epsilon_0\omega\sigma'/((\sigma')^2 + (\sigma'')^2)$, respectively. This formalism has the

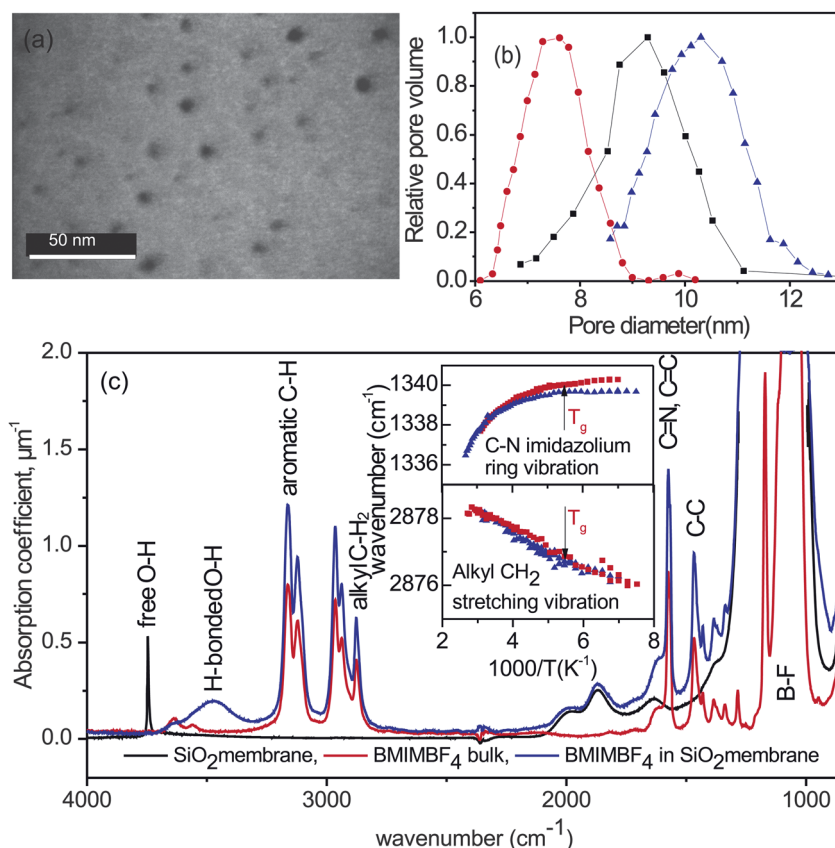


Fig. 1 (a) Scanning electron micrograph image showing the bottom view of SiO_2 membrane, (b) pore-size distribution obtained by NMR-cryporometry, and (c) infrared absorption spectra of bulk BMIM BF_4 , empty porous SiO_2 membrane and porous SiO_2 membrane filled with BMIM BF_4 . Inset: frequency of two characteristic vibrations of the imidazolium cation: (top) C–N imidazolium ring vibration and (bottom) CH_2 alkyl chain CH_2 asymmetric stretching. The temperature dependence of bulk (squares) and confined (triangles) ionic liquid is compared. The calorimetric glass transition temperature, T_g , of bulk BMIM BF_4 (184 K) is also indicated.

advantage of suppressing the contribution of the dc conductivity and electrode polarization to the dielectric spectra. However, a direct microscopic interpretation of the modulus spectra is not trivial. This makes it possible to compare the resulting diffusion coefficients to those obtained by PFG NMR in bulk and under conditions of geometrical confinement.

The real part of the complex conductivity function σ' in bulk ionic liquids is characterised on the intermediate frequency regime by a plateau, the value of which directly yields the dc conductivity, σ_0 , as well as the characteristic frequency, f_c , at which dispersion sets in and turns into a power law at higher frequencies. At lower frequencies, σ' decreases from σ_0 and this is due to electrode polarization.^{24,25} It is well established that charge transport in bulk ionic liquids is characterized by two main quantities obtained from the dielectric spectra, namely, σ_0 and f_c .^{21,23,24} However, σ' obtained for ionic liquids in nanopores exhibits weak frequency dependence in the spectral range assigned to σ_0 . This is attributed to contributions of a distribution of pores with different diameters.¹⁴ The typical conductivity and modulus spectra of the ionic liquid BMIM BF₄ in SiO₂ nanopores of mean pore diameter of 7.5 nm are shown in Fig. 2. In the current case, we assign σ_0 to the value of σ' at f_c . It is an experimental fact that the frequency corresponding to the peak in the imaginary part of the complex modulus is equivalent to f_c .^{21,26} The spectra in the whole temperature range investigated coincide upon scaling with respect to σ_0 and f_c as demonstrated in the insets of Fig. 2. This indicates identical thermal activation of the underlying mechanisms of charge transport and electrode polarization even under geometrical confinement.

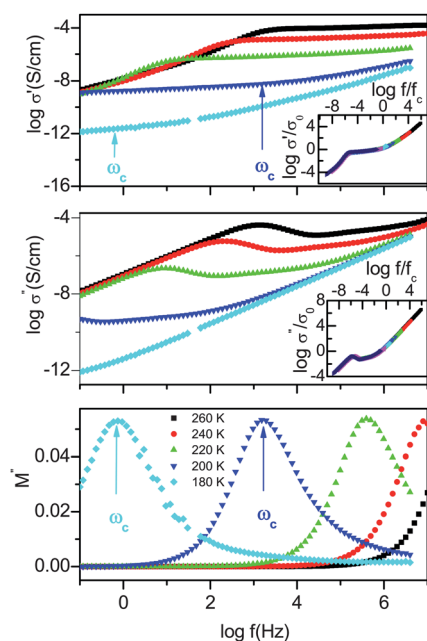


Fig. 2 Complex conductivity function ($\sigma'(\omega) = \sigma'(\omega) + i\sigma''(\omega)$) and the imaginary part of complex electrical modulus $M''(\omega)$ of the ionic liquid 1-butyl-3-methylimidazolium tetrafluoroborate—BMIM BF₄ versus frequency at different temperatures in 7.5 nm silica nanopores as indicated. Inset: scaling with respect to the characteristic frequency f_c and dc conductivity σ_0 . The error bars are comparable to the sizes of the symbols, if not specified otherwise.

The main transport quantities σ_0 and ω_c for the bulk ionic liquid exhibit Vogel–Fulcher–Tammann (VFT)-type of thermal activation as shown in Fig. 3. At higher temperatures, both σ_0 and ω_c obtained under nano-confinement coincide with the bulk values within the limits of experimental accuracy. However, systematic enhancement of the two quantities with decreasing pore diameters is found at lower temperatures. This is discussed in further detail later in the present article.

The Barton–Nakajima–Namikawa (BNN) relation, exemplified by plots of σ_0 versus ω_c ($=2\pi f_c$), for a variety of bulk ionic liquids was recently found to hold over timescales spanning more than ten orders of magnitude.²³ This was explained within the framework of Einstein and Einstein–Smoluchowski relations.^{27,28} Under conditions of geometrical confinement, additional parameters such as finite size effects may influence the measured macroscopic response. It is remarkable that the BNN relation also holds for confined ionic liquids in all the pore sizes reported in the current study (see the inset of Fig. 3). This indicates that the underlying mechanism(s) of charge transport remains unaltered even in nanopores.

Einstein and Smoluchowski proposed a microscopic description of diffusion enabling the determination of diffusion coefficients from dielectric spectra. Within this framework, the diffusion coefficient, D , is given by $D = (\lambda^2 \omega_c / 2)$, where λ denotes the mean ion jump length in the timescale of ω_c . Fig. 4 shows a comparison of diffusion coefficients from BDS and PFG NMR spanning more than 12 orders of magnitude. A value of $\lambda = 0.16$ nm was used to determine D from the dielectric spectra according to the procedure described in our previous articles.²⁹ At higher temperatures, the measured diffusion coefficients from BDS and PFG NMR coincide for both bulk and confined ionic liquids. Remarkable increases are observed in the low temperature regime under conditions of geometrical confinement. For nanopores with a mean diameter of 7.5 nm, the diffusion coefficients are more than two orders of magnitude higher than their corresponding bulk values (see Fig. 4). The ionic mobility, μ , can be determined using Einstein's relation: $\mu = (qD/kT)$, where k and q denote Boltzmann constant and elementary charge, respectively. Alternatively, the mobility can also be given by $\mu = (v/E)$, where v refers to the mean ionic velocity and E denotes the electric field. Since identical electric field is applied for all the measurements, the higher ionic mobility in small pores can be traced back to a corresponding increase in v . In contrast to hexafluorophosphate-based ionic liquid for which slower dynamics was observed and eliminated by

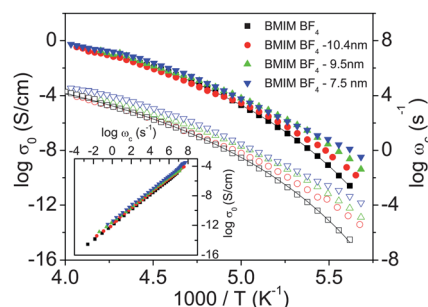


Fig. 3 Temperature dependence of dc-conductivity $\sigma_0(T)$ (open symbols) and characteristic rate of charge transport ω_c (filled symbols) for bulk and confined BMIM BF₄ in silica nanopores. Lines represent VFT fits of bulk data and the parameters are given in Table 1. Inset: dc-conductivity σ_0 versus ω_c for bulk and confined BMIM BF₄.

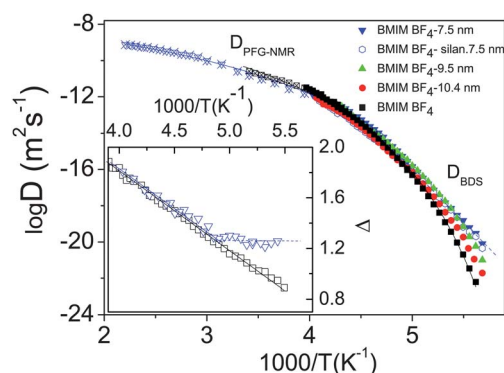


Fig. 4 Diffusion coefficients *versus* inverse temperature as obtained by PFG NMR (crossed open symbols) and BDS measurements (filled symbols) for bulk and confined BMIM BF₄ as indicated. The lines denote fits by VFT (continuous) and Arrhenius (dotted) equations. Inset: difference quotients $\Delta = [(\log D(1/T_i) - \log D(1/T_{i+1})) / (1/T_i - 1/T_{i+1})]^{-1/2}$; ($i = 1, \dots, 105$) *versus* $1/T$ for the bulk BMIM BF₄ (square symbols) and confined in porous silica with a mean diameter of 7.5 nm (denoted by triangles) as determined experimentally and $\Delta = [d \log(D)/d(1/T)]^{-1/2}$ as obtained from the above VFT (continuous lines)—and Arrhenius (represented by dotted lines)—fit functions.

pore-surface modification, the diffusion coefficients in the current IL remain unaltered by silanization.¹⁴ This indicates that the pore wall/IL interactions associated with adsorption and desorption processes only play a minimal role and cannot explain the present results.

The temperature dependence of bulk transport properties of ionic liquids is described by the empirical Vogel–Fulcher–Tammann (VFT) equation expressed as: $D(T) = D_\infty \exp[-B/(T - T_0)]$, where D_∞ , B , and T_0 denote the diffusion coefficient in the high temperature limit, a constant and Vogel temperature, respectively. These parameters are substituted accordingly for other transport quantities described in the current work (see Table 1). It is instructive to check how well the diffusion coefficient follows the VFT equation under confinement. The derivative technique provides a means of verifying the subtle features of the transport quantities spanning several orders of magnitude.^{26,30} Within this description, the VFT equation is given by $d \log \sigma_0 / d(1/T) = -B(1 - (T_0/T))^{-2} \log e$. When the resultant derivative of the diffusion coefficient is plotted against inverse temperature, the VFT equation is transformed into an Arrhenius-type relation (expressed as: $\log D(T) = \log D_\infty - (E_A/k_B T) \log e$, where E_A denotes the activation energy) with a single slope as demonstrated in the inset of Fig. 4. Under conditions of nanometric confinement, a change from a Vogel–Fulcher–Tammann into an Arrhenius-like thermal

activation is observed, resulting in an enhancement of diffusivities by more than two orders of magnitude.

The resultant macroscopic transport properties of materials in confining space are determined by a subtle interplay between surface- and confinement effects.^{26,31–33} Strong interactions between the host system and the guest molecules lead to slower dynamics while spatial restriction of certain molecules in nanometric length-scales may have the reverse effect of enhancing the molecular dynamics. Whereas the former can be quantitatively described by considering the strong guest–host interactions, the latter is often qualitatively explained in terms of the existence of a dynamic length-scale governing the glass transition, co-operatively re-arranging regions, among others.

In the current work, we attribute the increase of ionic mobility to changes in ion packing due to the reduction of the density of the ionic liquids in small pores. A simple explanation can be obtained by considering the problem of packing density of cylindrical balls in cylindrical containers. Both experimental and theoretical studies of spherical balls in cylindrical containers indicate that the mass density decreases by up to 7% when the radii of the balls become comparable to that of the confining cylinder. Neglecting surface effects, the mass density ρ_c under confinement can be expressed by the empirical relation: $\rho_c = \rho_b - A/R - B/H$, where ρ_b , R , and H denote the bulk mass density, radius of the cylinder, and height of the cylinder, respectively. A and B are geometrical constants. More rigorous approaches involving density functional theory and Voronoi–Delaunay methods deliver comparable results concerning the decrease of mass density under confinement.^{34–38} This approach has been extended to explain molecular properties with considerable success. Recently, Shi and Sorescu employed atomistic simulations to study diffusion in the ionic liquid HMIM NTf₂ confined in carbon nanotubes of diameter 4.5 nm and observed significant decrease in the mass density compared to the bulk value.¹⁵ As a result, the diffusion coefficients increased by about two orders of magnitude. Our present results confirm this picture at lower temperatures for BF₄-based ionic liquid (see Fig. 4). In a recent review article, it was pointed out that 2.1% change in the mass density of bulk poly(vinyl) acetate corresponds to variation of the structural relaxation rate over 8 orders of magnitude.³⁹ This underscores the role of density in determining the dynamic properties of glass-forming materials in the absence of strong IL–pore wall interactions. Molecular dynamics is thus controlled not only by thermal but also by density effects.^{40,41} It is remarkable that the dielectric properties of the ionic liquids based on tetrafluoroborate and hexafluorophosphate anions, which appear similar at first glance, are dominated by different mechanisms, namely, confinement and surface effects, respectively. Detailed molecular dynamics simulations are underway to unravel the origin of these differences as well as the key parameter defining the charge transport properties under nanometric confinement. Regarding the coinciding plots of diffusion coefficients at higher temperatures, we argue that due to the higher thermal energy of the molecules, the effects of density on dynamics are minimal in the high temperature regime. We expect that these effects should, however, show up in static quantities like the dielectric strength. Studies in this direction are in progress for low molecular-weight glass-forming liquids.

In conclusion, translational diffusion in 1-butyl-3-methylimidazolium tetrafluoroborate ionic liquid confined in non-intersecting silica nanopores with mean pore diameters down to 7.5 nm is studied by broadband dielectric spectroscopy and pulsed field gradient NMR. It is found that the diffusion coefficients are

Table 1 Fit parameters obtained from the VFT equation ($\omega(T) = \omega_\infty \exp[-B/(T - T_0)]$) for the characteristic rate, dc conductivity and diffusion coefficients of bulk BMIM BF₄ and BMIM BF₄ confined in 7.5 nm nanopores. The pre-exponential factor is re-named in accordance with the quantity under study

Transport quantity	$\log \omega_\infty$ (s ⁻¹)	B/K	T_0/K
ω_c	13.5 ± 0.2	1402 ± 20	140 ± 1
σ_0	1.6 ± 0.2	1352 ± 19	142 ± 2
D	-7 ± 0.3	1202 ± 25	146 ± 1
$D_{7.5 \text{ nm}}$	-7.6 ± 0.3	1167 ± 15	140 ± 1

enhanced by more than two decades in comparison to their bulk values. This is attributed to changes in ion packing under two-dimensional geometrical confinement, a view supported by recent atomistic simulations.

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